AQUAWATCH CENTRAL WATER QUALITY LABORATORY

ENVIRONMENTAL TESTING DIVISION * WATER QUALITY ANALYSIS & MONITORING SECTION

Testing Methodologies: APHA 23rd Edition / Bureau of Indian Standards (IS 3025 Baseline)

WATER QUALITY ANALYSIS REPORT

TEST REPORT OF WATER SAMPLE ANALYSIS AND QUALITY EVALUATION

1. SAMPLE, LOCATION & TESTING PARTICULARS

Report Number:	REP-20261105-596C09	**Sample Identifier:**	SMP-20261105-RL32
Water Body Name:	Mula-Mutha River - Pune	**Water Body Type:**	River
Sample Location:	Bund Garden, Pune, Pune	**GPS Coordinates:**	18.5204 N, 73.8567 E
Sample Matrix / Type:	Surface Water	**Sample Collection Date:**	05-Nov-2026 12:40 IST
Testing / Analysis Date:	05-Nov-2026 00:00 IST	**Report Generation Date:**	05-Nov-2026 18:14 IST
Testing Protocol / Standard:	Standard Methods (APHA / IS 3025)	**Technician / Analyst:**	LAB

2. WATER QUALITY INDEX (WQI) & BASELINE ASSESSMENT

Predicted Water Quality Index (WQI):	**41.52 / 100**	**Baseline Drinking Safety Screening:**	**HARMFUL**
Quality Grading / Classification:	Grade B - Good Quality (Suitable	**Regulatory Baseline Reference:**	CPCB Class C / BIS IS 10500:2012
Analytical Assessment Model:	Decision Tree	Screening Scope:	Baseline Physical-Chemical-Biological

3. MEASURED PARAMETERS & ANALYTICAL TEST RESULTS (14 PARAMETERS)

No.	Tested Parameter	Reference Test Method	Unit	Standard Baseline Limit	Measured	Status
1	**pH**	APHA 4500-H+ B / IS 3025 (Pt	pH Units	6.5 - 8.5	**8.40**	Within
2	**Dissolved Oxygen (DO)**	APHA 4500-O C / IS 3025 (Pt	mg/L	>= 4.0	**6.70**	Within
3	**Biochemical Oxygen Demand**	APHA 5210 B / IS 3025 (Pt 44	mg/L	<= 3.0	**3.20**	Exceeds
4	**Chemical Oxygen Demand (CO**	APHA 5220 B / IS 3025 (Pt 58	mg/L	<= 250.0	**16.00**	Within
5	**Turbidity**	Nephelometric / IS 3025 (Pt	NTU	<= 5.0	**1.00**	Within
6	**Total Dissolved Solids (TD**	Gravimetric / IS 3025 (Pt 16	mg/L	<= 500.0	**497.00**	Within
7	**Electrical Conductivity (E**	Conductivity Meter / IS 3025	uS/cm	<= 1500.0	**575.00**	Within
8	**Water Temperature**	Thermometric / IS 3025 (Pt 9	deg C	Ambient (15 - 35)	**28.00**	Within
9	**Nitrate (as NO3-)**	UV Spectrophotometric / IS 3	mg/L	<= 45.0	**0.50**	Within
10	**Phosphate (as PO4---)**	Stannous Chloride / APHA 450	mg/L	<= 0.10	**0.78**	Exceeds
11	**Chloride (as Cl-)**	Argentometric / IS 3025 (Pt	mg/L	<= 250.0	**58.48**	Within
12	**Total Coliform**	Multiple Tube Fermentation /	MPN/100 mL	<= 5000	**35.00**	Within
13	**Fecal Coliform**	Membrane Filtration / IS 162	MPN/100 mL	<= 500	**1.80**	Within
14	**Sulphate (as SO4--)**	Turbidimetric / IS 3025 (Pt	mg/L	<= 200.0	**72.36**	Within
15	**Ammonia Nitrogen (NH3-N)**	Phenate Method / APHA 4500-N	mg/L	<= 0.50	**0.43**	Within

4. OBSERVATIONS & RECOMMENDATIONS

- Threshold exceedances detected in key baseline parameters; sample is non-compliant for untreated drinking or direct potabil
- Recommended Treatment: Enhanced multi-stage treatment including coagulation, flocculation, advanced filtration, and robust
- Surveillance: Conduct root-cause sanitary inspection upstream and schedule repeat confirmatory sampling within 48 to 72 hou

5. VERIFICATION & AUTHORIZATION

Reporting Laboratory Analyst:

LAB

Designation: Authorized Laboratory Analyst / Environmental Chemist

Analysis Date: 05-Nov-2026 00:00 IST

Authorized Signatory Verification:

[Signature of Authorized Signatory]

Quality Assurance Officer / Lab In-Charge

6. REPORT AUTHENTICITY DETAILS & REGULATORY DISCLAIMER

1. AUTHENTICITY NOTICE: This document is an electronically generated water quality analytical report compiled from laboratory measureme
2. STATUTORY DISTINCTION: This electronically generated report does NOT constitute an officially accredited statutory test certificate
3. SCOPE OF RESULTS: Analytical test results relate solely to the sample as tested and submitted for the specified monitoring point. Th